E3 Fresnel diffraction

E3.1 Introduction

The principal aims of this experiment are (i) to test the theory of Fresnel diffraction for circular apertures and obstacles, straight edges and slits and (ii) to obtain familiarity with Fresnel half-period zones and the Cornu spiral. Effects on the optical axis can be studied quantitatively, whereas off-axis effects may be studied semi-quantitatively.

Before starting, please review the notes on laser safety at the start of the lab manual.

References

- There are many excellent textbooks on waves and optics, among which 'Optics' by E. Hecht is particularly useful.
- The Handout supporting your lecture course 'Oscillations, Waves and Optics' covers relevant background material. Diffraction is covered from page H135. Fresnel Diffraction is covered from page H211. The Cornu spiral appears on page H215.

E3.1.1 Notes on the apparatus

This experiment uses a similar optical bench to that provided for the Fraunhofer experiment, E2, and which has already been used in the experiments in E1. You may wish to refer to the introductory description of that experiment to familiarise yourself with this apparatus. You are provided with an aperture plate to create the diffraction patterns and several lenses for magnifying them — an observing screen, a small circular aperture, a slit of variable width which can also serve as a straight edge, and some circular obstacles.

E3.1.2 Theory and hints on your experimental investigations

Fresnel diffraction from a circular aperture is normally considered in terms of Fresnel half-period zones (H229). The nth Fresnel zone is the annular region of the aperture between radii of $\sqrt{(n-1)\lambda R}$ and $\sqrt{n\lambda R}$, where R is the distance from the aperture to observing screen (assuming plane-wave illumination), or $\frac{1}{R} = \frac{1}{r_a} + \frac{1}{r_b}$ for distances r_a and r_b on either side of the zoneplate (H235) otherwise. Central zones give contributions of the same magnitude, but alternating sign, to the amplitude-phase diagram. For outer zones, the contributions are decreased by the obliquity factor and any spatial incoherence of the incident radiation. The intensity on-axis can be calculated mathematically, but intensities off axis are much harder to handle computationally. Circular obstacles give rise to the phenomenon known as Poisson's spot, with diffraction fringes around the spot and at the edge of the shadow.

For straight edges and slits, Fresnel diffraction is normally understood in terms of the Cornu spiral (Fig. E3.0), which is simply another amplitude-phase diagram. The distance measured *along* the spiral is a scaled version of the distance measured across the aperture. The origin corresponds to the point in the aperture plane that intercepts the shortest (i.e. line-of-sight) distance from source to observer. Both qualitative and quantitative predictions can be made using the spiral.

As part of your investigations, you may wish to start by checking whether for the dimensions of the apparatus and bench length you intend to use, you expect to see Fraunhofer or Fresnel diffraction. Fresnel conditions are obtained by moving the observing screen towards the aperture. This would normally lead to the production of very small diffraction patterns, so in practice a lens is used to project the Fresnel pattern onto a distant screen. You can use standard formulae for the locations of the conjugate planes of the lens and for the image magnification, and should assume that any lens aberrations are unimportant. It is useful to remember that the diffraction pattern, which is the 'object' in your optical set-up, is formed in space near the diffracting aperture.

E3.2 Procedure

The procedure described here is the same as that given for the Fraunhofer experiment. The aim is to align the beam accurately to the optical rail.

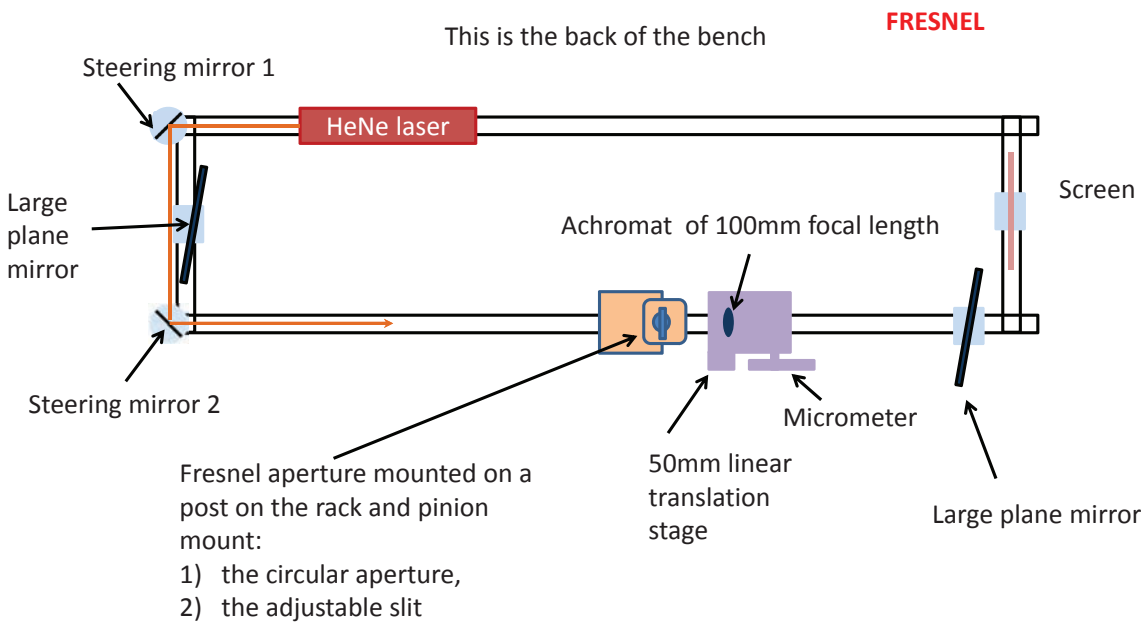


Figure E3.1: Illustration of component layout. The rack-and-pinion mount, and linear stage are mounted in the right-hand portion of the bench in order to facilitate close observation of the diffraction pattern on the screen while still being able to reach the adjustments with the left hand.

E3.2.1 Set up the laser

Check that the laser itself is delivering a beam at approximately 24cm above the table top, and if need be, move the laser up or down in the postholders until it is horizontal and at the right height. Check that the postholders are tight and the mounts secured to the bench.

Check that the front and back plates of the mirror mount of Steering Mirror 1 are at this point nearly parallel – otherwise you will run out of adjustment range later. Check that the postholder base is clamped to the bench.

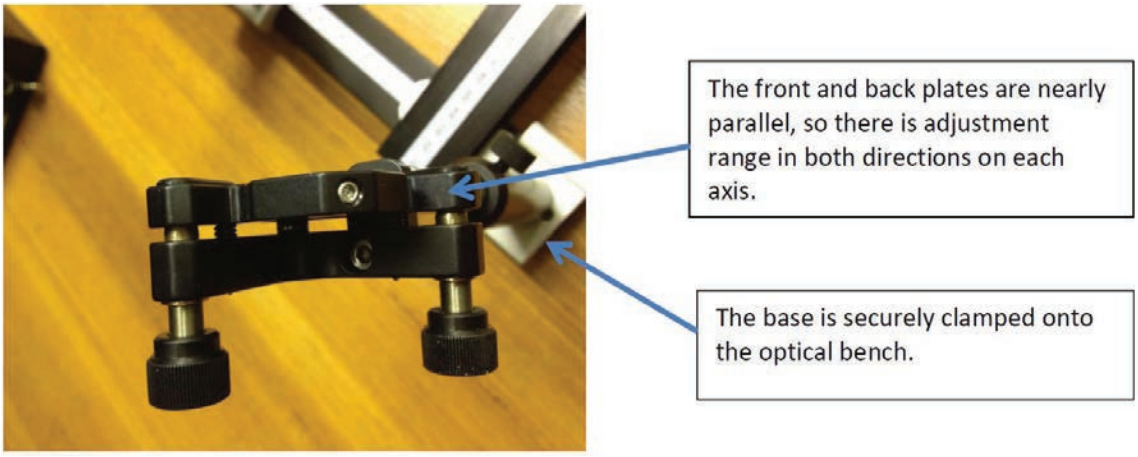


Figure E3.2: Correct positioning of corner mirror.

E3.2.2 Set up Steering Mirror 1

Loosen the post-clamping screw of Steering Mirror 1 so the mount can be raised so that the beam falls nearly centrally on the mirror. It is important never to attempt to rotate any post-mounted component without first loosening the clamping screw in the postholder. Setting this to be 'finger-tight' permits rotation without the post sliding down in the holder.

With the post loosened, set the height and crude rotation of the mirror so that it deflects the beam through a right angle to Steering Mirror 2. The entire mirror mount and post can be rotated until the spot on Steering Mirror 2 is nearly on the central axis. Then clamp the post. Use the vertical adjustment screw of Steering Mirror 1 and a ruler to ensure that the beam stays at around 24cm height as it propagates to Steering Mirror 2.

E3.2.3 Setup Steering Mirror 2

Check that the front and back plates of the mirror mount of Steering Mirror 2 are nearly parallel. Check that the postholder base is clamped to the bench. Unclamp the post of Steering Mirror 2 and set its height so that the beam hits it centrally, and rotate it so that it deflects the beam close to centrally along the bench. Clamp the postholder.

E3.2.4 Optical alignment of the laser beam to the target

Go back to Steering Mirror 1 and use the fine control screws to centre the beam on Steering Mirror 2 – use the set screw on the post's axis as a guide. Place the metal

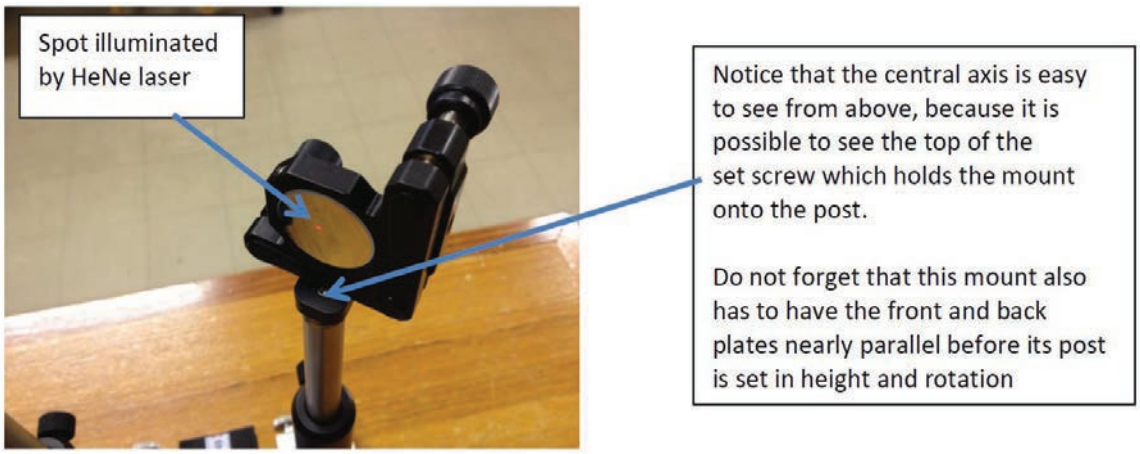


Figure E3.3: Spot location on corner mirror.

pointer in the right kind of postholder (see above) and set it to around 24cm height above the table. Place it as close to Steering Mirror 2 as it will fit on the main longitudinal optical rail, push it to the rail so that the fixed side of the saddle lies in contact with the rail, and centre the beam on the point using Steering Mirror 1. Move the postholder assembly as far to the right-hand end of the rail as possible, and recentre the beam, but now using Steering Mirror 2. Iterate until there is no further improvement – if you followed the earlier instructions this should not need more than 2 or maybe 3 iterations.

Quick check: if the procedure has been followed correctly the achromat of 100mm focal length can be positioned in the postholder on the linear stage so that the laser beam passes exactly through its centre.

E3.2.5 Alignment of the beam to the screen

Without any lens or aperture, use the first large plane mirror to deflect the beam to the centre of the second large plane mirror – as ever, only rotate the mirrors when the post is not clamped! The second large plane mirror can then be rotated so as to send the beam to a central point on the screen. Heights of both these mirrors and the screen should be set so that the beam is roughly central vertically as well as horizontally, and then all three items can be clamped. Their bases should, of course, also be firmly secured to the relevant rails.

E3.2.6 Location of the target

For the experiment, first place the metal-mounted circular aperture in the rack-and-pinion mount in the orientation shown in Fig E3.1(the postholder should be positioned to the right). The micrometer drive on the linear stage should initially be set to 45mm in order to allow accurate movement of the lens away from the aperture. Using the metric scale on the optical rail, the lateral position of the aperture should be set so that

it is approximately 100mm to the left of the lens position (when the micrometer is set to 45mm on its scale).

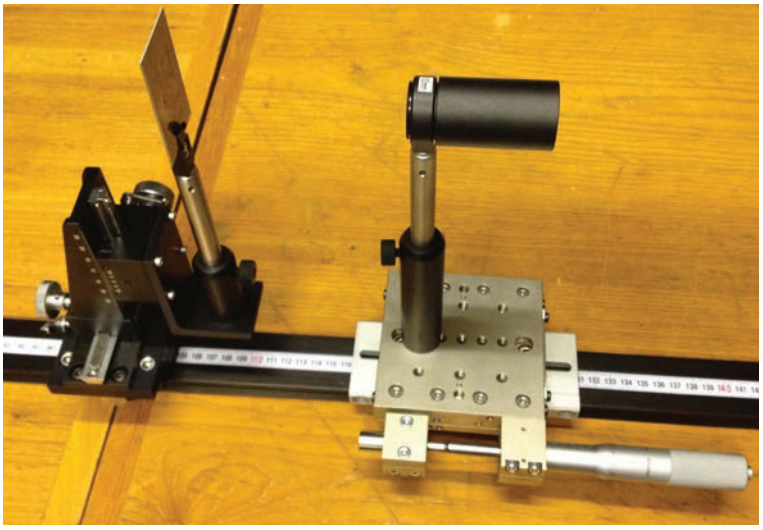
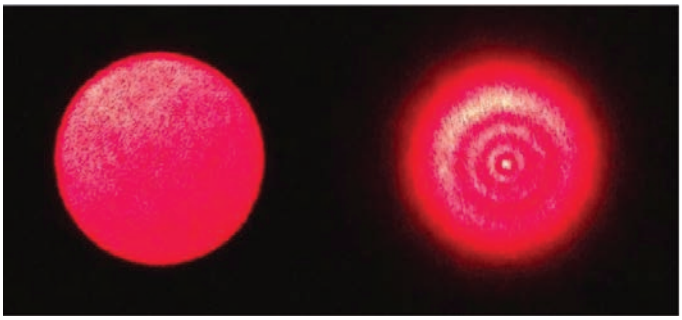


Figure E3.4: The arrangement of the apparatus for imaging the Fresnel pattern.

The adjustments of the rack-and-pinion mount can be used to centre the aperture in the laser beam, and a beautiful, circularly-symmetric Airy pattern should be visible on the screen (as expected from Fraunhofer diffraction at distances for away from the aperture). Make sure the rack-and-pinion mount is secured to the rail. After this, the lens can be inserted into the mount of the linear stage, in order to form an image of the diffraction pattern near the aperture.



The left-hand figure shows the focused image of a clean circular aperture; the right-hand figure shows the Fresnel diffraction pattern obtained by displacing the lens away from the aperture plane.

Figure E3.5: The left-hand picture shows the image of a clean, circular aperture. The right-hand image shows the Fresnel diffraction pattern obtained by displacing the lens of 100mm focal length away from the initial, imaging, position.

E3.3 Fresnel diffraction experiments

From lectures, and from your supervision work, you should now be familiar with diffraction. You should aim to test the theory of Fresnel diffraction as quantitatively as you

can; this should be possible for both circular apertures and straight edges. Specific experiments, feasible with the apparatus in the laboratory, are discussed in the following.

Tips on setting up the optics for your experiment

To find the distance from your aperture to the diffraction pattern, first arrange a lens to obtain a sharp, magnified image of the diffracting aperture on the screen. If you then move the lens by a known distance, z, away from the aperture you may assume — under any experimental conditions you are likely to be using — that z is the distance from the aperture of the diffraction pattern of which a magnified image is formed on the observing screen. You are provided with a linear translation stage which allows smooth movement and position measurement to a precision of about 10 μm. You should not use this hint without thinking carefully about it and demonstrating you understand it in your lab notebook!

E3.4 Quantitative analysis of circular apertures: on-axis behaviour

For circular apertures you should quantitatively test the expectations of Fresnel theory for the on-axis diffraction pattern, for instance by measuring the distances along the axis at which the intensity is zero. You should consider carefully what you expect to observe on-axis as the number of zones across the aperture changes. For a fixed aperture size you can alter the number of zones across the aperture by investigating the pattern at different distances from the aperture.

E3.5 Semi-quantitative analysis of circular apertures: off-axis behaviour

For circular apertures expectations of Fresnel theory for the off-axis diffraction pattern can be tested semi-quantitatively, for instance by measuring the fringe widths. The separation of the fringes can be understood in terms of Fresnel zones.

E3.6 Qualitative analysis of circular obstacles

Qualitatively test the expectation of the Fresnel theory for a circular obstacle. Discuss the limiting case of the "screen" very close to the obstacle.

E3.7 Quantitative analyses of straight edges

For straight edges you should quantitatively test the expectations of Fresnel theory for the diffraction pattern, for instance by measuring the separation of the fringes corresponding to the intensity minima, exploiting the Cornu spiral.

E3.8 Qualitative analysis of slits and limiting cases

By using the variable-width slit, qualitatively investigate the observed diffraction pattern in relation to the expectations of Fresnel theory. Discuss in particular the extreme cases of very wide and very narrow slit, with the aid of the Cornu spiral. What is the relation with Fraunhofer diffraction, if any?

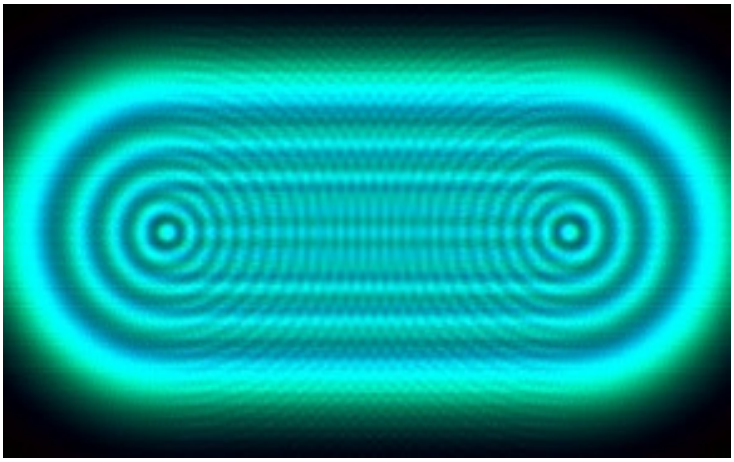


Figure E3.6: The Fresnel diffraction pattern of a 'race-track' aperture. Image courtesy of Dauger Research Inc.